

Literature-Already-Answered Packet
Carbery's Schatten almost-orthogonality question
Status: literature already answered, not a new automatic-discovery proof

Source and later answer

The source paper is:

Anthony Carbery, *Almost-orthogonality in the Schatten–von Neumann classes*,
Journal of Operator Theory 62(1):151–158, 2009.

The local copy is included as `source_paper.pdf`.

The later supporting paper is:

arXiv:2606.04116, Ziang Chen, Paata Ivanisvili, Jose Madrid, and Haozhu Wang,
Carbery's inequality in the Schatten–von Neumann classes.

The local copy is included as `supporting_paper_2606.04116.pdf`.

Open question location

The following full-width crop is taken from article page 106, PDF page 7 of Carbery's paper. It shows Section 3.3 and equation (3.1), where Carbery asks for the sharpened estimate with the ℓ_2 -operator norm of $(\alpha_{jk}^{p'})$.

3.2. Is it possible to formulate a meaningful question or result in the case $1 < p < 2$?

3.3. The statement of Theorem 1.3 does not formally recover the Cotlar–Stein Lemma in the limiting case $p = \infty$. At the other endpoint $p = 2$, estimate (2.1) is inefficient, and so neither does the statement of the theorem recover the discussion of the case $p = 2$ (see (1.1)) from the Introduction.

In fact, the Cotlar–Stein Lemma and the discussion of the case $p = 2$ (see (1.1)) actually suggest (via naïve “interpolation”) that an alternative result may be possible. Under the hypotheses of Theorem 1.3, let A_p be the l^2 -operator norm of $(\alpha_{jk}^{p'})$. Do we have

$$(3.1) \quad \left\| \sum_j T_j \right\|_p \leq A_p^{1/p'} \left(\sum_j \|T_j\|_p^p \right)^{1/p}?$$

(Note that $A_p \leq A$ since the Schur norm is dominated by the l^2 -operator norm and since we may assume via Hölder’s inequality that each $\alpha_{jk} \leq 1$.) We do not know, even in the commutative case, whether (3.1) holds, even when p is an integer. As an easier question, what about (3.1) with the Schur norm of $(\alpha_{jk}^{p'})$ replacing the operator norm?

3.4. Applying the commutative version of Theorem 1.3 in the situation when there are only two summands we obtain an L^p inequality: if $f, g \in L^p$ and $p \in \mathbb{N}$, $p \geq 2$, then

Question

In Carbery’s notation, under the hypotheses of Theorem 1.3, let A_p be the ℓ_2 -operator norm of the matrix $(\alpha_{jk}^{p'})$, where $p' = p/(p-1)$. He asks whether

$$\left\| \sum_j T_j \right\|_p \leq A_p^{1/p'} \left(\sum_j \|T_j\|_p^p \right)^{1/p}. \quad (C)$$

The surrounding text says that this was unknown even in the commutative case and even when p is an integer.

Supporting answer

The supporting paper explicitly identifies this same question and states that it answers it affirmatively. The following crop is from page 3 of arXiv:2606.04116.

and combines the resulting multilinear expression with a Schur-type test. Similar inequalities for many functions have been studied also in [4].

In this paper we affirmatively answer the question posed by Carbery in [2, Section 3.3, (3.1)] in full generality for the Schatten–von Neumann classes. We prove the following sharp statement, valid for *all* real $p \geq 2$ and for *arbitrary* (not necessarily positive or self-adjoint) operators in S_p .

Theorem 1.1. *Let $N \geq 2$, $p \geq 2$, and let $T_1, \dots, T_N \in S_p(\mathcal{H})$. Let $(\alpha_{ij})_{1 \leq i, j \leq N}$ be a symmetric matrix with entries $\alpha_{ij} \in [0, 1]$ satisfying (1). Then*

$$\left\| \sum_{j=1}^N T_j \right\|_{S_p} \leq \|(\alpha_{ij}^{p'})\|_2^{1/p'} \left(\sum_{j=1}^N \|T_j\|_{S_p}^p \right)^{1/p}.$$

Remark 1.2. Since

$$\|(\alpha_{ij}^{p'})\|_2 \leq \sqrt{\|(\alpha_{ij}^{p'})\|_1 \|(\alpha_{ij}^{p'})\|_\infty} = \|(\alpha_{ij}^{p'})\|_1 = \sup_i \sum_j \alpha_{ij}^{p'} \leq \|(\alpha_{ij})\|_1,$$

Theorem 1.1 improves Carbery’s even-integer Schur- (α_{ij}) estimate, answers the ℓ_2 -operator-norm question (2), and it extends [5, Theorem 1.1] to Schatten classes and all $p \geq 2$.

Remark 1.3. The exponent p' in the constant $\|(\alpha_{ij}^{p'})\|_2$ is optimal, i.e., the counter-example from [5] shows that no larger exponent than p' is admissible on the coefficients α_{ij} .

Theorem (Chen–Ivanisvili–Madrid–Wang, Theorem 1.1). *Let $N \geq 2$, $p \geq 2$, and let $T_1, \dots, T_N \in S_p(H)$. Let (α_{ij}) be a symmetric matrix with entries in $[0, 1]$ satisfying the same almost-orthogonality hypotheses as in Carbery’s question. Then*

$$\left\| \sum_{j=1}^N T_j \right\|_{S_p} \leq \|(\alpha_{ij}^{p'})\|_2^{1/p'} \left(\sum_{j=1}^N \|T_j\|_{S_p}^p \right)^{1/p}.$$

Claim. *Carbery’s Section 3.3 question (3.1) is already answered affirmatively by arXiv:2606.04116.*

Proof. The match is direct. Carbery’s C_p and the supporting paper’s S_p are the Schatten–von Neumann classes with the Schatten p -norm. Carbery asks whether estimate (C) holds with A_p equal to the ℓ_2 -operator norm of $(\alpha_{jk}^{p'})$. The supporting paper says it answers the question posed by Carbery in “[2, Section 3.3, (3.1)]” and proves exactly this operator-norm estimate for all real $p \geq 2$, for arbitrary operators in $S_p(H)$. The finite-family theorem is the finite form of the question, and the same paper records the countable-family passage in its concluding remarks.

Therefore the exact open question shown in the source crop has an explicit later affirmative answer. $\square$

Classification

This packet belongs in `solutions/literature_already_answered/`. The supporting paper knows it is answering Carbery’s Section 3.3 question and cites the original source. This is therefore a retrieval/status result, not a new proof or new counterexample from the smoke-test run.

Verification notes

- The source statement is not a same-paper false positive: Carbery leaves (3.1) open in the 2009 paper.
- The supporting source is separate and later, and explicitly cites Carbery’s Section 3.3, equation (3.1).
- The hypotheses and norm in the displayed theorem match the question: both use the almost-orthogonality coefficients α_{ij} , the Schatten norm, and the ℓ_2 -operator norm of $(\alpha_{ij}^{p'})$.
- This packet does not claim to settle Carbery’s distinct commutative Section 3.5 question.

Search bounds and provenance

The original paper was located after a broader bibliographic search for Carbery’s 2009 title. OpenAlex metadata identified the Journal of Operator Theory article and an Edinburgh Research Explorer repository PDF. The supporting answer is the arXiv paper 2606.04116 already found in the run’s paper cache. Text search in both PDFs located the original Section 3.3 question and the supporting paper’s explicit answer passage.

Reproducibility

The screenshot crops were produced from the included PDFs by rendering the relevant pages to PNG in `tmp/` and running

`code/crop_evidence.py`.

The final rendered packet is `solution_packet.pdf`; build intermediates are kept in `tmp/`.